

Properties of Trapezoids and Kites

Jonathan R. Bacolod

A **trapezoid** is a quadrilateral with one pair of **parallel sides**.

Bases and Legs

The parallel sides are called the **bases**, while the nonparallel sides are called the **legs**.

Isosceles Trapezoid

In an **isosceles trapezoid**, the legs are congruent, the base angles are congruent, and the diagonals are congruent.

Midsegment of a Trapezoid

The **midsegment** of a trapezoid is parallel to the bases and has length

$$\frac{b_1 + b_2}{2},$$

where b_1 and b_2 are the lengths of the bases.

Kite

A **kite** is a quadrilateral with two distinct pairs of adjacent congruent sides.

Diagonals of a Kite

The diagonals of a kite are **perpendicular**.

$$d_1 \perp d_2$$

Perpendicular Bisector

One diagonal of a kite is the **perpendicular bisector** of the other diagonal.

Examples

Example 1

The midsegment of a trapezoid measures 18 cm. One base is 24 cm. Find the other base.

$$18 = \frac{24 + b}{2}$$

$$36 = 24 + b$$

$$b = 12$$

The other base is 12 cm.

Example 2

The diagonals of an isosceles trapezoid are $3x + 5$ and $5x - 11$. Find x .

$$3x + 5 = 5x - 11$$

$$16 = 2x$$

$$x = 8$$

Therefore, $\boxed{x = 8}$.

Example 3

A kite has adjacent sides measuring 9 cm, 9 cm, 14 cm, and 14 cm. Explain why the figure is a kite.

The sides form two pairs of adjacent congruent sides:

$$9 = 9$$

and

$$14 = 14$$

Therefore, the figure is a **kite**.

Example 4

One pair of opposite angles of a kite measures 74° . Find the measure of the congruent opposite angle.

Opposite angles are congruent.

$$\boxed{74^\circ}$$

The congruent opposite angle measures 74° .

Example 5

The diagonals of a kite measure 16 cm and 24 cm. Find the area.

$$A = \frac{1}{2}d_1d_2$$

$$A = \frac{1}{2}(16)(24)$$

$$A = 192$$

The area is $\boxed{192 \text{ cm}^2}$.

Practice Exercises

Practice 1

The midsegment of a trapezoid measures 18 cm. One base is 24 cm. Find the other base.

$$18 = \frac{24 + b}{2}$$

Practice 2

The diagonals of an isosceles trapezoid are $3x + 5$ and $5x - 11$. Find x .

$$3x + 5 = 5x - 11$$

Practice 3

A kite has adjacent sides measuring 9 cm, 9 cm, 14 cm, and 14 cm. Explain why the figure is a kite.

Practice 4

One pair of opposite angles of a kite measures 74° . Find the measure of the congruent opposite angle.

Practice 5

The diagonals of a kite measure 16 cm and 24 cm. Find the area of the kite.

$$A = \frac{1}{2}d_1d_2$$

Activities

Activity 1

A trapezoid has bases measuring 14 cm and 26 cm. Find the length of its midsegment.

$$m = \frac{14 + 26}{2}$$

Activity 2

In an isosceles trapezoid, one base angle measures 72° . Find the other three angles.

Activity 3

In trapezoid $PQRS$, the bases are parallel. If $\angle P = 118^\circ$, find $\angle S$.

Activity 4

The diagonals of a kite intersect at right angles. One diagonal bisects the other. Identify the property demonstrated.

Activity 5

In kite $JKLM$, diagonal JL bisects diagonal KM . If $KM = 26$ cm, find the length of each half of KM .

$$\frac{26}{2} = 13$$

Thank you!